

TECHNO-ECONOMIC OPTIMIZATION OF AN OFF-GRID PV-DIESEL-BESS MICROGRID

HOMER Pro Engineering Study

Project Item	Value
Project	Remote Facility PV-Diesel-BESS
Software	HOMER Pro 3.11.2
Study type	Off-grid hybrid microgrid techno-economic optimization
Primary load	120 kWh/day; 43,800 kWh/year
Baseline diesel price	\$1.20/L
Baseline solar resource	5.38 kWh/m ² /day
Project lifetime	25 years
Nominal discount / inflation	8% / 2%

Prepared as an engineering portfolio study

1. Executive Summary

This study evaluates an off-grid hybrid power system supplying a remote facility using solar photovoltaic generation, a diesel generator, lithium-ion battery energy storage, and a bidirectional system converter. HOMER Pro was used to optimize component sizing and dispatch while enforcing zero annual capacity shortage.

For the baseline case (diesel fuel price \$1.20/L and solar resource 5.38 kWh/m²/day), the least-cost architecture consists of 44.5 kW of PV, a 25 kW fixed-capacity diesel generator, 62 kWh of nominal lithium-ion storage, and a 19.3 kW converter under HOMER Cycle Charging dispatch. The solution has a total net present cost (NPC) of \$166,159.40, a levelized cost of energy (COE) of approximately \$0.2935/kWh, and annual operating cost of about \$5,359.

Key KPI	Baseline result
PV capacity	44.5 kW
Diesel generator	25.0 kW
Battery storage	62 kWh nominal; 49.6 kWh usable
Converter	19.3 kW
Renewable fraction	83.3%
Diesel consumption	2,269 L/year
Generator operation	334 h/year
PV production	69,929 kWh/year
Generator production	7,303 kWh/year
Excess electricity	29,446 kWh/year
Unmet load / capacity shortage	0 / 0
Total NPC	\$166,159.40
Levelized COE	\$0.2935/kWh

The sensitivity analysis shows that higher diesel prices favor greater renewable penetration and lower diesel use, while stronger solar resources reduce the cost of energy and can reduce the PV capacity required to achieve an economical solution.

2. Project Objective and Scope

The objective is to identify an economically attractive and technically feasible off-grid microgrid configuration for a remote facility. The study focuses on energy adequacy, component sizing, dispatch behavior, lifecycle economics, fuel dependence, and sensitivity to fuel price and solar resource.

- Supply the annual AC primary load without unmet energy.
- Optimize PV, battery and converter sizing while retaining a 25 kW diesel generator option.
- Compare dispatch behavior and select the least-cost feasible architecture.
- Quantify NPC, COE, annual operating cost, renewable fraction and diesel consumption.
- Evaluate sensitivity to diesel price and solar global horizontal irradiation (GHI).

3. System Description and Assumptions



System Architecture

Component	Name	Size	Unit
Generator	Generic 25kW Fixed Capacity Genset	25.0	kW
PV	Generic flat plate PV	44.5	kW
Storage	Generic 1kWh Li-Ion	62	strings
System converter	System Converter	19.3	kW
Dispatch strategy	HOMER Cycle Charging		

Schematic

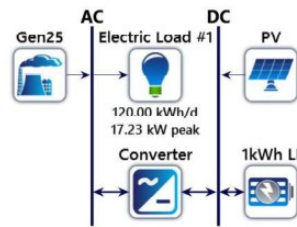


Figure 1. HOMER Pro system architecture for the optimized baseline case.

Parameter	Assumption / value
AC load	120 kWh/day average; approximately 17.23 kW peak
Annual AC energy	43,800 kWh/year

Solar GHI baseline	5.38 kWh/m ² /day
PV	Generic flat plate, DC bus
PV derating factor	80%
PV cost basis	\$1,000/kW capital; \$800/kW replacement; \$10/kW-year O&M
Generator	Generic 25 kW fixed-capacity diesel genset
Generator minimum load	25% (6.25 kW)
Diesel price baseline	\$1.20/L
Battery	Generic 1 kWh Li-ion units; 90% round-trip efficiency
Battery minimum SOC	20%
Converter	Generic system converter; 95% inverter and rectifier efficiency
Dispatch	HOMER Cycle Charging selected in optimum
Annual capacity shortage	0%
Nominal discount rate	8%
Expected inflation	2%
Real discount rate	5.88%
Project life	25 years

The solar resource values used in this training study are assumed monthly averages rather than measured site data. Accordingly, the results should be interpreted as a portfolio-grade techno-economic demonstration rather than a bankable design.

4. Optimized Baseline Architecture

HOMER selected a renewable-dominant architecture with diesel generation serving primarily as firm backup and seasonal support. The battery provides energy shifting between periods of PV surplus and periods of low or zero solar production.

Component	Optimized size	Engineering role
PV array	44.5 kW	Primary annual energy source
Diesel genset	25.0 kW	Firm backup and battery-support generation
Li-ion BESS	62 × 1 kWh	Short-duration energy shifting and reserve
Converter	19.3 kW	DC/AC and AC/DC power conversion
Controller	Cycle Charging	Coordinates generator operation and storage charging

5. Electrical Energy Balance



Cash Flow

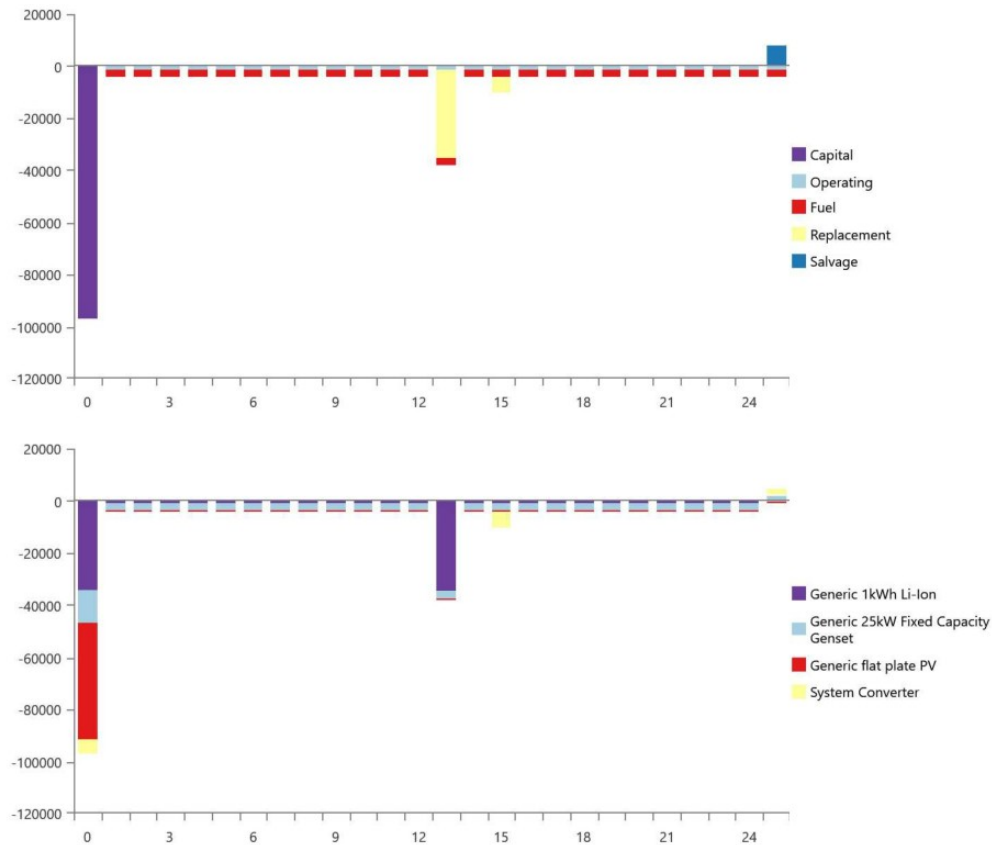


Figure 2. HOMER electrical summary for the baseline optimum.

Energy quantity	Value	Comment
PV production	69,929 kWh/year	90.5% of gross electrical production
Generator production	7,303 kWh/year	9.46% of gross electrical production

Total generation	77,232 kWh/year	Before storage/converter losses and curtailment
AC primary load	43,800 kWh/year	100% served
Excess electricity	29,446 kWh/year	38.1% of gross production
Unmet electric load	0 kWh/year	Reliability target satisfied
Capacity shortage	0 kWh/year	Reliability target satisfied

The high excess-electricity value is a significant design observation. The PV array is economically attractive because it displaces diesel fuel, but the facility cannot consume or store all midday and seasonal surplus. In a real project, flexible loads, water pumping, thermal storage, EV charging, hydrogen production, export capability, or alternative storage sizing could be evaluated to improve utilization.

6. Diesel Generator Performance



Electrical Summary

Excess and Unmet

Quantity	Value	Units
Excess Electricity	29,446	kWh/yr
Unmet Electric Load	0	kWh/yr
Capacity Shortage	0	kWh/yr

Production Summary

Component	Production (kWh/yr)	Percent
Generic flat plate PV	69,929	90.5
Generic 25kW Fixed Capacity Genset	7,303	9.46
Total	77,232	100

Consumption Summary

Component	Consumption (kWh/yr)	Percent
AC Primary Load	43,800	100
DC Primary Load	0	0
Total	43,800	100

Figure 3. Diesel generator annual operating pattern and performance.

Generator metric	Result
Electrical production	7,303 kWh/year
Hours of operation	334 h/year

Number of starts	140 starts/year
Mean electrical output	21.9 kW
Minimum electrical output	6.25 kW
Maximum electrical output	25.0 kW
Capacity factor	3.33%
Fuel consumption	2,269 L/year
Specific fuel consumption	0.311 L/kWh
Mean electrical efficiency	32.7%

The generator operates only 334 hours per year, confirming that it functions mainly as backup rather than as the primary energy source. Its mean output of 21.9 kW while operating is relatively high compared with its 25 kW rating, which is consistent with Cycle Charging: when the generator is committed, it can operate at a productive loading level while surplus power charges the battery.

7. PV Performance



Generator: Generic 25kW Fixed Capacity Genset (Diesel)

Generic 25kW Fixed Capacity Genset Electrical Summary

Quantity	Value	Units
Electrical Production	7,303	kWh/yr
Mean Electrical Output	21.9	kW
Minimum Electrical Output	6.25	kW
Maximum Electrical Output	25.0	kW

Generic 25kW Fixed Capacity Genset Fuel Summary

Quantity	Value	Units
Fuel Consumption	2,269	L
Specific Fuel Consumption	0.311	L/kWh
Fuel Energy Input	22,330	kWh/yr
Mean Electrical Efficiency	32.7	%

Generic 25kW Fixed Capacity Genset Statistics

Quantity	Value	Units
Hours of Operation	334	hrs/yr
Number of Starts	140	starts/yr
Operational Life	44.9	yr
Capacity Factor	3.33	%
Fixed Generation Cost	2.57	\$/hr
Marginal Generation Cost	0.328	\$/kWh

Generic 25kW Fixed Capacity Genset Output (kW)

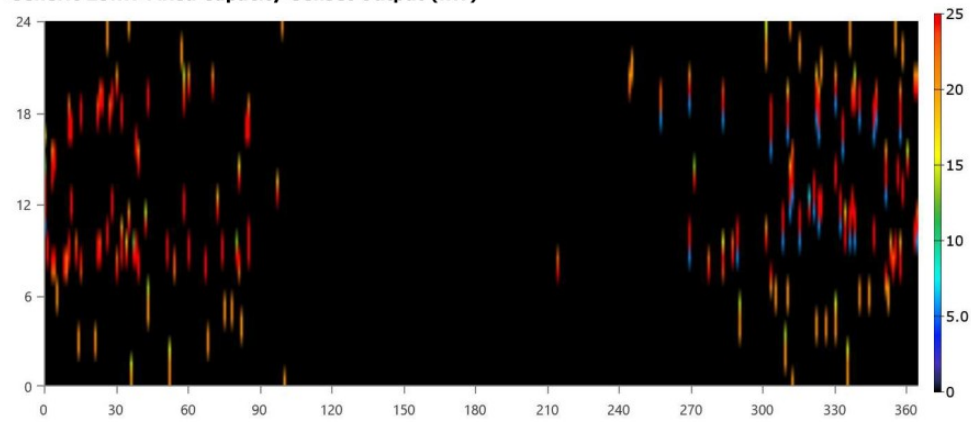


Figure 4. Annual PV power-output distribution.

PV metric	Result
Rated capacity	44.5 kW
Mean output	7.98 kW

Mean daily production	192 kWh/day
Maximum output	42.1 kW
Capacity factor	17.9%
Annual production	69,929 kWh/year
PV penetration	160%
PV levelized cost	\$0.0556/kWh

PV penetration above 100% does not mean that the load is supplied by PV at every instant. It indicates that annual PV production is large relative to annual load. The mismatch between solar availability and load demand explains why storage, diesel backup, and significant excess electricity remain present.

8. Battery Energy Storage Performance



PV: Generic flat plate PV

Generic flat plate PV Electrical Summary

Quantity	Value	Units
Minimum Output	0	kW
Maximum Output	42.1	kW
PV Penetration	160	%
Hours of Operation	4,380	hrs/yr
Levelized Cost	0.0556	\$/kWh

Generic flat plate PV Statistics

Quantity	Value	Units
Rated Capacity	44.5	kW
Mean Output	7.98	kW
Mean Output	192	kWh/d
Capacity Factor	17.9	%
Total Production	69,929	kWh/yr

Generic flat plate PV Output (kW)

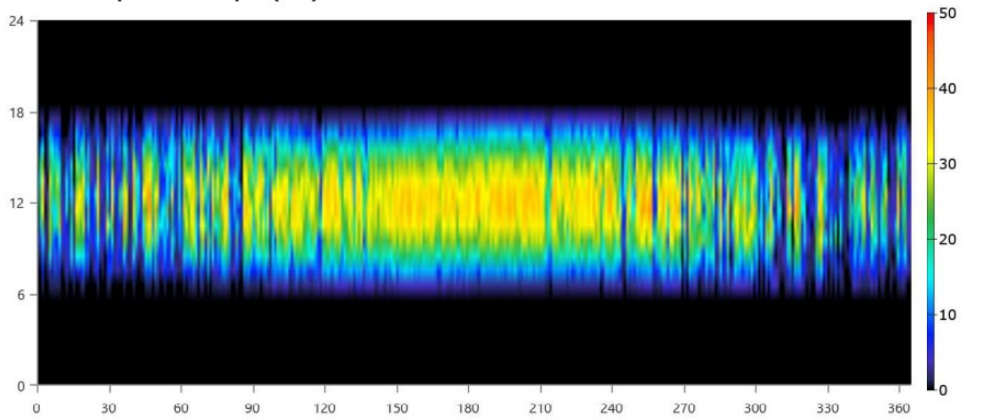


Figure 5. Battery state-of-charge distribution and annual statistics.

Battery metric	Result
Number of battery units	62
Nominal capacity	62.0 kWh

Usable nominal capacity	49.6 kWh
Autonomy	9.92 h
Energy in	15,513 kWh/year
Energy out	13,977 kWh/year
Losses	1,552 kWh/year
Annual throughput	14,733 kWh/year
Expected life	12.6 years
Lifetime throughput	186,000 kWh

The 62 kWh battery provides approximately 9.9 hours of autonomy under HOMER's definition. Its expected life of 12.6 years is shorter than the 25-year project life, so replacement cost materially contributes to lifecycle economics. The state-of-charge plot shows regular cycling throughout the year.

9. Converter Performance



Storage: Generic 1kWh Li-Ion

Generic 1kWh Li-Ion Properties

Quantity	Value	Units
Batteries	62.0	qty.
String Size	1.00	batteries
Strings in Parallel	62.0	strings
Bus Voltage	6.00	V

Generic 1kWh Li-Ion Result Data

Quantity	Value	Units
Average Energy Cost	0.144	\$/kWh
Energy In	15,513	kWh/yr
Energy Out	13,977	kWh/yr
Storage Depletion	16.4	kWh/yr
Losses	1,552	kWh/yr
Annual Throughput	14,733	kWh/yr

Generic 1kWh Li-Ion Statistics

Quantity	Value	Units
Autonomy	9.92	hr
Storage Wear Cost	0.193	\$/kWh
Nominal Capacity	62.0	kWh
Usable Nominal Capacity	49.6	kWh
Lifetime Throughput	186,000	kWh
Expected Life	12.6	yr

Generic 1kWh Li-Ion State of Charge (%)

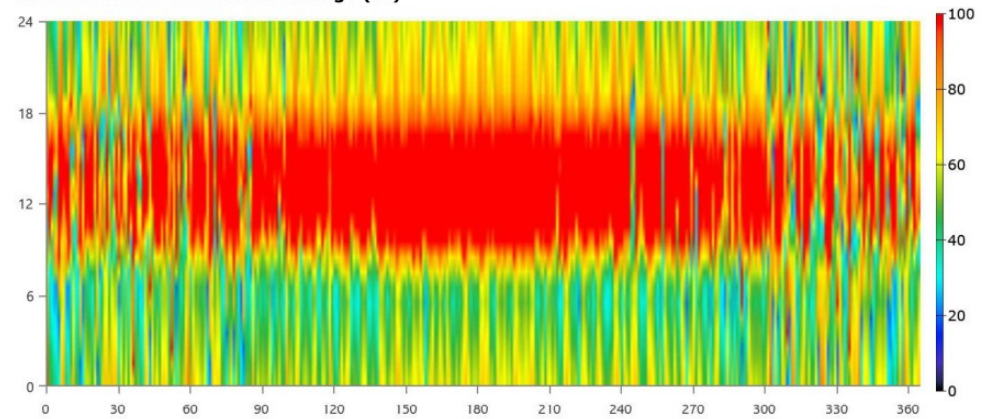


Figure 6. Inverter and rectifier annual operating patterns.

Converter metric	Inverter	Rectifier
Capacity	19.3 kW	19.3 kW
Mean output	4.75 kW	0.559 kW

Maximum output	17.2 kW	19.3 kW
Hours of operation	8,455 h/year	305 h/year
Energy out	41,653 kWh/year	4,898 kWh/year
Energy in	43,845 kWh/year	5,156 kWh/year
Losses	2,192 kWh/year	258 kWh/year

The inverter operates for most of the year because the PV and battery are connected to the DC side while the primary load is AC. Rectifier operation is much less frequent and is associated mainly with AC-side generator energy being transferred to the DC bus for battery charging.

10. Economic Analysis

Economic metric	Baseline result
Initial capital	\$96,877
Total NPC	\$166,159.40
Levelized COE	\$0.2935/kWh
Annual operating cost	\$5,359/year
Project lifetime	25 years
Real discount rate	5.88%

NPC represents the present value of all project lifecycle costs after discounting future cash flows, including capital, replacements, operation and maintenance, fuel, and salvage effects. COE expresses the lifecycle cost on a per-unit-energy basis, allowing different architectures to be compared in \$/kWh.

NPC contribution	Approx. present value
Li-ion storage	\$58,526
Diesel generator	\$49,614
PV array	\$50,253
System converter	\$7,767
Total system	\$166,159

The lifecycle cost is distributed across PV, storage and diesel generation rather than being dominated by a single component. The generator has modest initial capital but incurs fuel and operating costs, while the battery has substantial capital and replacement exposure.

11. Fuel Use, Emissions and Renewable Contribution



Converter: System Converter

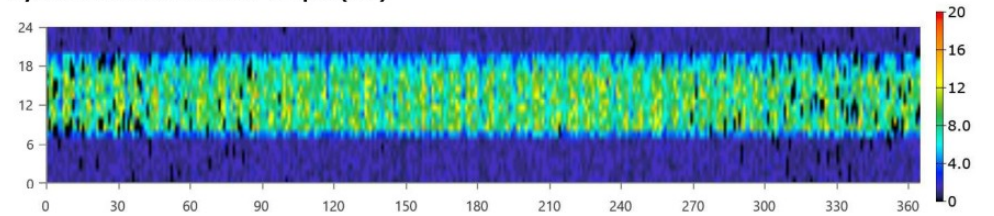
System Converter Electrical Summary

Quantity	Value	Units
Hours of Operation	8,455	hrs/yr
Energy Out	41,653	kWh/yr
Energy In	43,845	kWh/yr
Losses	2,192	kWh/yr

System Converter Statistics

Quantity	Value	Units
Capacity	19.3	kW
Mean Output	4.75	kW
Minimum Output	0	kW
Maximum Output	17.2	kW
Capacity Factor	24.7	%

System Converter Inverter Output (kW)



System Converter Rectifier Output (kW)

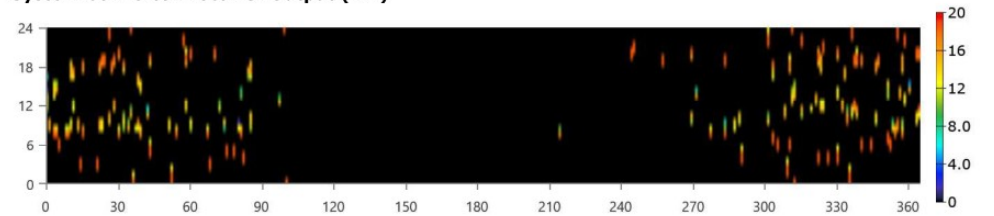


Figure 7. Diesel fuel consumption and modeled emissions.

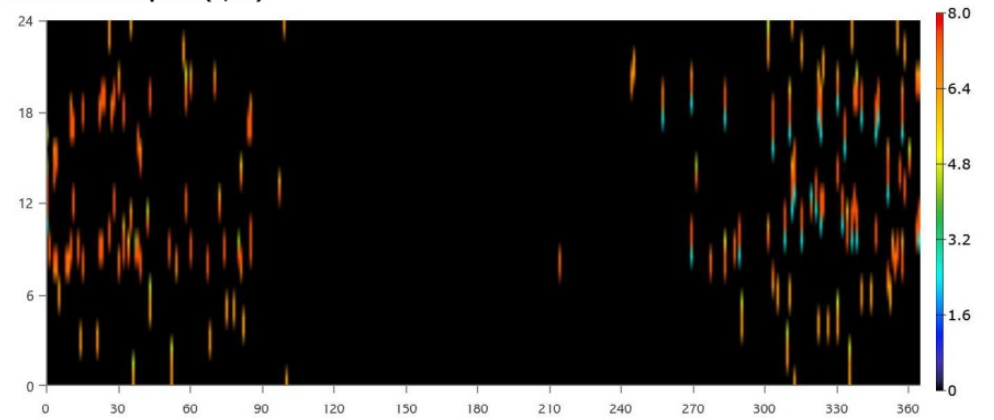


Fuel Summary

Diesel Consumption Statistics

Quantity	Value	Units
Total fuel consumed	2,269	L
Avg fuel per day	6.22	L/day
Avg fuel per hour	0.259	L/hour

Diesel Consumption (L/hr)



Emissions

Pollutant	Quantity	Unit
Carbon Dioxide	5,941	kg/yr
Carbon Monoxide	37.1	kg/yr
Unburned Hydrocarbons	1.63	kg/yr
Particulate Matter	0.222	kg/yr
Sulfur Dioxide	14.5	kg/yr
Nitrogen Oxides	34.9	kg/yr

Figure 8. HOMER renewable contribution summary.

Metric	Result
Renewable fraction	83.3%
Capacity-based renewable fraction	64.0%
Diesel consumption	2,269 L/year

Average fuel use	6.22 L/day
Average fuel use per operating hour	6.79 L/hour
CO2 emissions	5,941 kg/year
CO emissions	3.64 kg/year
Unburned hydrocarbons	0.991 kg/year
Particulate matter	0.222 kg/year
SO2 emissions	14.5 kg/year
NOx emissions	34.9 kg/year

The 83.3% renewable fraction demonstrates that most of the load-related energy requirement is supplied from renewable generation after accounting for system interactions. The remaining diesel use provides reliability during periods of inadequate solar generation and storage availability.

12. Sensitivity Analysis

Two sensitivity variables were evaluated: diesel fuel price (\$0.80, \$1.20 and \$1.60/L) and solar GHI (4.50, 5.38 and 6.00 kWh/m²/day). HOMER re-optimized the architecture for each of the nine combinations.

Diesel \$/L	GHI	PV kW	BESS kWh	Conv. kW	Ren. %	Fuel L/yr	NPC	COE \$/kWh
0.80	4.50	44.2	64	20.5	80.4	2,646	\$161,475	\$0.285
0.80	5.38	38.5	63	19.8	81.2	2,533	\$152,726	\$0.270
0.80	6.00	38.3	58	18.6	81.8	2,466	\$148,500	\$0.262
1.20	4.50	44.1	67	19.7	81.0	2,570	\$175,202	\$0.309
1.20	5.38	44.5	62	19.3	83.3	2,269	\$166,159	\$0.293
1.20	6.00	40.0	62	19.5	83.4	2,261	\$160,989	\$0.284
1.60	4.50	51.8	63	20.8	83.0	2,302	\$188,074	\$0.332
1.60	5.38	54.4	69	16.6	87.6	1,704	\$177,512	\$0.314
1.60	6.00	49.2	70	16.7	87.8	1,671	\$171,292	\$0.303

The sensitivity results show the expected economic trade-off. Increasing diesel price encourages HOMER to invest in additional renewable generation and storage, reducing annual fuel consumption and increasing renewable fraction. Improving solar resource lowers lifecycle cost because each installed kilowatt of PV produces more useful energy.

For example, at the baseline GHI of 5.38 kWh/m²/day, increasing diesel price from \$0.80/L to \$1.60/L increases the optimized PV capacity from 38.5 kW to 54.4 kW, raises renewable fraction from 81.2% to 87.6%, and reduces diesel consumption from 2,533 to 1,704 L/year. At a fixed diesel price of \$1.20/L, increasing GHI from 4.50 to 6.00 kWh/m²/day reduces NPC from \$175,202 to \$160,989 and COE from \$0.309 to \$0.284/kWh.

All nine reported sensitivity optima use Cycle Charging, indicating that this dispatch strategy remains economically preferred across the tested fuel-price and solar-resource ranges.

13. Engineering Discussion

- **Reliability:** the optimized baseline serves the full annual load with zero unmet electric load and zero capacity shortage.
- **Renewable dominance:** PV supplies 90.5% of gross annual electrical production, while diesel generation is limited to 334 operating hours per year.
- **Curtailement opportunity:** 29,446 kWh/year of excess electricity is large relative to the 43,800 kWh/year load and should be investigated in a real design.
- **Storage utilization:** the BESS is actively cycled and provides approximately 9.9 hours of autonomy, but its expected 12.6-year life implies at least one lifecycle replacement.
- **Generator utilization:** Cycle Charging allows the genset to operate at relatively high output when committed, avoiding prolonged inefficient low-load operation.
- **Sensitivity:** fuel-price risk materially changes the optimal renewable investment, demonstrating why sensitivity analysis is essential for remote diesel-dependent systems.

A bankable engineering design would require measured or validated hourly load data, site-specific irradiance, temperature effects, equipment vendor quotations, battery degradation and warranty limits, generator maintenance schedules, fuel logistics, electrical protection, grounding, short-circuit studies, and detailed constructability assessment.

14. Conclusions and Recommendations

The HOMER Pro optimization demonstrates that a PV-diesel-battery hybrid system can supply the modeled remote facility reliably while substantially reducing diesel dependence. The baseline optimum uses 44.5 kW PV, a 25 kW diesel generator, 62 kWh of lithium-ion storage and a 19.3 kW converter. It achieves an 83.3% renewable fraction with only 2,269 L/year of diesel consumption, at an NPC of approximately \$166,159 and COE of \$0.2935/kWh.

The sensitivity analysis confirms that higher diesel prices make greater PV and storage investment economically attractive, while improved solar resource reduces lifecycle cost. For future project development, the highest-value next investigation is the large excess-energy fraction. Flexible demand or productive-use loads could improve utilization of the PV asset and potentially change the optimal battery and generator sizing.

- Replace assumed solar data with validated site-specific hourly resource data.
- Replace synthetic load with measured hourly facility demand where available.
- Obtain vendor-based PV, BESS, converter and generator costs.
- Evaluate flexible loads to absorb midday excess energy.
- Perform additional sensitivities for battery cost, PV cost, load growth and fuel escalation.
- For detailed design, follow the techno-economic study with electrical network, protection and interconnection studies.

15. Source and Study Limitations

Baseline component performance, economics, energy balance, fuel use, emissions and renewable metrics in this report are derived from the supplied HOMER Pro System Simulation Report for the baseline case. The nine-case sensitivity table is based on the HOMER sensitivity results captured during the study. The study is educational/portfolio-oriented and does not constitute a detailed design, tender specification, or investment-grade feasibility study.